

Problem-Solution Fit Canvas

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

Confirms the proposed solution addresses the root causes, behaviors, and limitations of the target user, filled out directly from the built system rather than as a speculative exercise.

1. CUSTOMER SEGMENT(S) [CS] Students engaged in self-paced or independent study (school, college, or online-course learners) who hit difficulty without a live instructor present at that moment.	6. CUSTOMER LIMITATIONS [CL] - Free-tier tools only, no dedicated tutor budget. - Browser-based access via laptop or phone. - Limited to whatever help is available at the moment of difficulty (e.g. late at night).	5. AVAILABLE SOLUTIONS PROS & CONS [AS] - Generic chatbots: always available, but tone-deaf to whether the student is frustrated, bored, or curious. - Static FAQs / re-reading material: free, but doesn't adapt or diagnose why the student is stuck. - Peer / instructor office hours: real understanding, but not available in the moment.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] - Repeatedly hitting the same wall with no adaptive support — happens in effectively every self-study session. - Generic help that reads the same regardless of whether the student is Confused, Frustrated, Bored, Curious, or Confident.	9. PROBLEM ROOT / CAUSE [RC] Existing tools respond to the content of a question, not the emotional state behind it, so support "sounds the same" for a struggling student and a bored one.	7. BEHAVIOR + ITS INTENSITY [BE] - Retries the same approach without changing strategy when stuck. - Disengages — skips ahead or multitasks — when bored. - Reflected in the training data itself: Frustrated and Bored are among the largest observed classes.
3. TRIGGERS TO ACT [TR] - A wrong answer on an unfamiliar concept. - A re-read that still doesn't click. - A solved problem with no immediate way to confirm mastery.	10. YOUR SOLUTION [SL] - A dual-model (BiLSTM + BERT) classifier reads the student's own description of the problem. - A data-driven decision engine decides how much to trust that read (HIGH / MEDIUM / LOW).	8. CHANNELS OF BEHAVIOR [CH] - ONLINE: Browser-based Streamlit app (local development + public Streamlit Cloud deployment). - OFFLINE: Not applicable — the tool is fully online, no offline channel exists.

	• An LLM generates a response tuned to both the detected emotion and the field of study.	
4. EMOTIONS BEFORE / AFTER [EM] • Before: frustrated, confused, or bored, and unsupported in the moment. • After (target): reassured and re-engaged, with a visible, honest trust signal instead of blind confidence in an automated read.		